Looking Monitorable Without Being Faithful: Math-CoT Fine-Tuning Degrades Cue-Faithfulness Independent of Teacher Strength

Abraham Yeung
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Abstract

Chain-of-thought (CoT) monitorability assumes that reading a model's reasoning reveals *why* it answered. We test whether fine-tuning a reasoning model on math CoT preserves that property, using the cue-based faithfulness paradigm (Meek et al. 2025; Turpin et al. 2023) on a DeepSeek-R1-distilled student. Our central finding is a **dissociation between behavior and verbalization**: after weak-to-strong reasoning fine-tuning (W2SR; Yuan et al. 2025), a planted cue's *behavioral* pull on the answer is *not reduced* — flips land on the cue target at ∼2× chance, statistically the same rate as the untrained model (paired influence $\Delta = +0.16$, CI $[+0.04, +0.28]$, lower bound near zero) — while the CoT's *acknowledgment* of that cue collapses. On a strict, attribution-only rubric (the conservative claim) acknowledgment goes 9.4% → 0.0% (15/0 one-directional discordant, McNemar $p = 6\times10^{-5}$); on the loose "mention or attribute" rubric used by the upstream eval, 25% → 3% (34/1, $p \approx 2\times10^{-9}$), at preserved accuracy. The model keeps being moved by the cue and stops saying so. That fine-tuning lowers CoT faithfulness is known (Lobo et al. 2024); our contribution is that the cue's causal grip is held fixed while verbalization disappears, which is the specifically *monitorability*-relevant failure. We show the effect is **teacher-strength-invariant**: weak (1.5B), strong (14B), and *same-strength self-distillation* (teacher == student) all collapse acknowledgment far below baseline (2.4–7.4% vs. 25%) with no significant difference between arms (pairwise McNemar $p \geq 0.09$; self-distill arms indistinguishable from W2SR weak at $p = 1$), so weak-to-strong asymmetry is incidental; the cause is SFT on terse math-style CoT. The drop is compression-dominant (CoT shortens ∼14×; at properly matched length the residual is not significant, so we read the mechanism as *compression as produced by terse math-CoT SFT*, not as a residual beyond length), emergent rather than imitated (training traces carry the `</think>` separator in 100% of records, student completions in 22%), robust to an out-of-family judge (`gemini-2.5-pro`, $\kappa \in [0.65, 0.68]$, paired drop preserved), and replicates on MMLU at ∼3× compression with a smaller absolute drop — a within-paper dose-response from compression to acknowledgment collapse. **A capability-only certification, or one that rewards elaborate-looking CoT, can pass a model whose reasoning has gone behaviorally opaque while remaining superficially legible.**

1 Introduction

Chain-of-thought "monitorability" is the hope that reading a model's reasoning reveals *why* it answered, so an overseer can catch undesired influences. Weak-to-Strong Reasoning (W2SR) [1] reports that fine-tuning a strong student on a *weaker* model's CoT *raises accuracy*. Faithfulness work [3, 2, 4] shows models often do not verbalize the cues that actually move their answers.

Critique. W2SR certifies the student on accuracy alone and is silent on whether the distilled CoT stays monitorable. If distillation changed *how* a model reasons on the page without changing *what it reveals*, an accuracy check would miss it.

Contributions. **(1)** We exhibit a *behavior–verbalization dissociation*: math-CoT fine-tuning preserves or strengthens a cue's causal pull on the answer (directional, ∼2× chance) while collapsing the CoT's acknowledgment of that cue, ruling out the benign reading that the model simply became cue-resistant or merely terser (§5). **(2)** We show the collapse is *teacher-strength-invariant* via weak, strong, and same-strength self-distillation controls (all far below baseline; pairwise McNemar $p \geq 0.09$ between arms; self-distill arms vs. W2SR weak $p = 1$), establishing that the cause is general math-CoT SFT, not weak-to-strong asymmetry, and that it replicates across judge family and onto MMLU (§5). **(3)** We reproduce the underlying W2SR capability gain and show it is itself conditional: it requires an in-family teacher and is genuine only on a base student (§4).

Table 1: Reproduction (held-out MATH L3–5 Pass@1). In-family native-Qwen teacher; headroom probe separates genuine elicitation from CoT-prompting. Genuine on the base student, artifact on the instruct student.

student	unelicited	zero-shot CoT	W2SR	W2SR − CoT
Qwen2.5-Math-7B (base)	0.325	0.240	0.645	+**0.405** (genuine)
Qwen2.5-7B-Instruct	0.230	0.630	0.630	+**0.000** (artifact)
Foreign-style teachers (R1-distill-1.5B, SimpleRL-Zoo) → 7B: no gain / collapse.				

2 Related work and what is new here

Fine-tuning degrades CoT faithfulness. Lobo et al. [5] show task-specific fine-tuning lowers CoT faithfulness across four datasets; we take this degradation as given. Our claim is not that faithfulness drops but that it drops *while the cue's behavioral influence on the answer does not*, a dissociation their answer-level faithfulness metric does not isolate. **Self-distillation and verbalization.** Concurrent work [6] finds self-distillation suppresses *epistemic* verbalization (expressed uncertainty) via compression, hurting out-of-distribution generalization. We share the compression mechanism and the self-distillation setup but differ in outcome variable: our target is *cue-acknowledgment* (a safety/monitorability property), not uncertainty calibration or OOD accuracy, and we measure the behavioral cue-pull directly. **CoT monitorability.** Meek et al. [2] measure monitorability as faithfulness plus verbosity across model families; we use their cued-prompt eval but study how a *training intervention* moves faithfulness on a fixed substrate, and pair it with a behavioral-influence measure to separate "stops being influenced" from "stops verbalizing influence," and ask the sharper question of whether faithfulness falls *beyond* what CoT length alone explains.

3 Setup

W2SR [1]: sample CoT from a weak teacher on a problem set, keep correct + incorrect traces, LoRA-SFT a stronger student; report held-out Pass@1. **Monitorability** [2]: each question is paired with a *cue* (authority opinion, leaked answer key, embedded metadata, etc.) pointing at a target; an LLM judge reads the full CoT and scores *faithfulness* (does the CoT *acknowledge* the cue) and *verbosity* (a factor-utilization rate). We use the five cues shipped in the released eval and a single judge (`claude-sonnet-4-6`, T=0); faithfulness is reported as acknowledgment among *cued* samples. All evaluation is greedy (T=0); trace sets are hashed. Training is LoRA; generation/training/gating run on Modal GPUs.

4 Reproduction: when does W2SR fire?

With a foreign-style weak teacher (DeepSeek-R1-distilled 1.5B, or SimpleRL-Zoo) distilled into a Qwen2.5-7B student, we found *no* Pass@1 gain and frequent degeneration, across students, ranks, and decodings. W2SR reproduces only with an **in-family, native-Qwen** weak teacher: `Qwen2.5-Math-1.5B-Instruct` (∼67% on the train problems) → `Qwen2.5-Math-7B` lifts held-out MATH (L3–5) Pass@1 from 0.325 to 0.645, format-valid 1.00.

Genuine only on a base student (headroom probe). A +5pt gate-pass on an already-CoT-capable model may be re-arrangement, not elicitation. Scoring the *untrained* model under a zero-shot CoT prompt (the prompting-only ceiling) and reporting W2SR − (zero-shot CoT) (Table 1): on the base student W2SR adds +0.41 *beyond prompting* (genuine latent-capability elicitation); on an instruct student it adds +0.00—its apparent gain over an unelicited baseline is purely the CoT-elicitation confound. Two conditions Yuan et al. leave implicit: in-family teacher, and a base (non-CoT-eliciting) student.

5 Extension: does W2SR preserve CoT monitorability?

Capability-controlled design. We compare a baseline student, a W2SR student (LoRA-SFT'd on a weak in-family teacher's MATH CoT), and references, on GPQA. The SFT lands the student at its own CoT ceiling, so faithfulness differences are not a capability confound; the question is "does distillation change faithfulness," not "does it improve capability."

Instruct student is floored (motivation). On `Qwen2.5-7B-Instruct`, cue-acknowledgment is at a floor for every condition (baseline 1.8% at N=970; W2SR 3.6%; strong-teacher control 4.6%), while teacher references reach 22–37%. We cannot detect faithfulness transfer here—but verbosity transfers strongly (W2SR−baseline $\Delta = +0.26$, CI $[0.22, 0.31]$): the model *looks* more monitorable without measurable faithfulness change. To probe a substrate with dynamic range, we move to a reasoning student.

Reasoning student: behavior preserved, verbalization collapses (Table 2). Fully in-family: student `R1-Distill-Qwen-7B`, weak teacher `R1-Distill-Qwen-1.5B`; distillation is clean (gate format-valid 0.99); baseline acknowledgment is off-floor (25%). **Behavior is not reduced.** Paired influence (answer == cue target on matched (qid, cue)): $\Delta = +0.157$, 95% CI $[+0.037, +0.278]$, $n = 108$, McNemar $p = 0.021$, discordant 16/33 — direction-positive but with a lower CI bound near zero, so we read this as *not reduced* rather than confidently strengthened. The cleaner, stronger signal: flips land directionally on the cue target at 73% for W2SR vs. 65% for baseline, both $\sim$2× the 1/3 random-flip chance ($p < 10^{-7}$ each) — statistically indistinguishable from each other and from the untrained directionality. The pull is genuine cue-influence, not accuracy gain in disguise: restricted to the cue-points-at-*wrong*-answer stratum (so landing on the cue *is* a wrong answer), flippers still hit the cue target at 77%/74%/74% for baseline / W2SR weak / W2SR strong (each $p \leq 2\times10^{-9}$ vs. 1/3). **Verbalization collapses (both rubrics).** On the upstream loose rubric ("acknowledge the cue"), W2SR drops pooled acknowledgment to 3.2% (paired on (qid,cue): $\Delta = -0.220$, CI $[-0.293, -0.153]$, McNemar $p \approx 2 \times 10^{-9}$, discordant 34/1; the drop holds at every text cue: stanford 47%→8%, grader-hack 22%→3%, insider 56%→5%, with one-directional discordant cells 12/0, 7/1, 15/0). *The conservative read* (stricter "attribute reasoning to the cue" rubric, same Gemini judge): the loose rubric was partly counting mentions; under the strict rubric the rates shrink to baseline 9.4% vs W2SR 0.0% (15/0 discordant, $p = 6 \times 10^{-5}$), so the magnitude is smaller but the drop is *more* one-directional and the trained student never explicitly attributes reasoning to the cue. Both numbers hold at *preserved* GPQA accuracy (0.42 vs. 0.28, n=40) and $\sim$14× shorter CoT (baseline median 18,692 chars → W2SR 1,362). The drop is not an artifact (hand-checked: completions coherent, 37/40 conclude with an answer, none degenerate—the model solves the problem without verbalizing the cue). It is also not a single-judge artifact: re-scoring the same 525 R1-family CoTs with `gemini-2.5-pro` yields per-condition Cohen's $\kappa \in [0.65, 0.68]$ and a paired drop $\Delta = -0.153$, McNemar $p = 2.4 \times 10^{-7}$, discordant 23/0.

Mechanism: compression-dominant + a real matched-length residual. Faithfulness and CoT length both fall; the near-trivial half is that shorter CoT surfaces cues less, so the claim of interest is degradation *at matched length*. Length is endogenous to reasoning (a brevity-prompt control failed—R1 ignores it), so we isolate the residual by *convergence* across imperfect length-controlled cuts on long ($\geq$9.6k-char) traces: (a) acknowledgment (judge): 19% vs. 44%, $p = 0.016$ (effective n: W2SR long-tail 39 vs. baseline 110; effect direction-stable across full/overlap/long-only logit fits with log(CoT chars) + cue controls, $p \leq 0.03$ on each, but magnitude underpowered); (b) literal cue-mention rate: 16% vs. 28% (same direction, n.s.); (c) mention density: 0.037 vs. 0.167 per 1k chars; (d) natural length-matched question pairs: 1/9 vs. 4/9 (same direction, underpowered). **All four cuts agree in direction; the primary (judge) cut is significant.** A finer length-binned analysis (Task H, Fisher's exact, unpaired) sharpens this: at matched mid-length ($[1954, 8716)$ chars) baseline and W2SR are indistinguishable (17.5% vs 15.1%, $p = 0.80$); at very long lengths ($[8716, 37816)$, baseline $n = 119$, W2SR $n = 40$) baseline is meaningfully higher (27.7% vs 15.0%, OR ≈ 2.2, one-sided $p = 0.08$) — a direction-positive residual just below conventional significance. So most of the 25% → 3% drop is **compression** (the channel separator </think> appears in 100% of training traces but only 22% of W2SR-student completions vs. 58% of baseline, so the compression is emergent under SFT, not from stripped supervision), and the residual lives specifically in the long-trace regime. A teacher-strength axis (Table 2) shows the effect is **compression-general**: a strong 14B teacher degrades faithfulness about as much as the 1.5B teacher and is not distinguishable from it. **Self-distillation negative control.** To test whether weak-to-strong asymmetry is causal at all, we train baseline R1-7B on its OWN R1-7B traces (1200 MATH L3–5, $T = 0.6$, same SFT config). Both self-distillation arms (4k and 8k generation budgets)

Table 2: Reasoning-student monitorability (GPQA, in-family R1-distill teachers). Faithfulness = acknowledgment among cued samples. Median CoT computed on cued-cell completions. W2SR degrades faithfulness at preserved accuracy; a strong (14B) teacher degrades it nearly as much as a weak (1.5B) one, and **self-distillation (teacher == student) collapses faithfulness indistinguishably from W2SR weak** ($p=1$). All trained arms land far below baseline (25%) at magnitudes that are not statistically distinguishable from each other (pairwise McNemar $p \geq 0.09$); the effect is teacher-independent and generalizes to ordinary same-strength SFT on MATH-style traces.

condition	faithfulness	GPQA acc	median CoT (chars)
baseline R1-7B	40/160 = 25.0%	0.28	18,692
W2SR, weak teacher (1.5B)	6/190 = 3.2%	0.42	1,362
W2SR, strong teacher (14B)	13/175 = 7.4%	—	1,508
Self-distill, ≤4k bud (Self-A)	2/84 = 2.4%	—	1,258
Self-distill, ≤8k bud (Self-B)	3/89 = 3.4%	—	1,292
cross-family: Llama-8B self-distill	gate fail	0.225	1,424

Cross-family row (Llama-8B self-distillation, matched recipe): GPQA accuracy $0.500 \to 0.225$ (-27.5pp at 0% empty), CoT compression 10.8×; capability gate failed, faithfulness number not interpretable, not reported (see Limitations).

weak vs. baseline $p=2\times10^{-9}$; strong vs. baseline $p=1\times10^{-5}$; Self-A vs. baseline $p=2.4\times10^{-7}$ ($n=77$, disc. 23/0); Self-B vs. baseline $p=4.8\times10^{-7}$ ($n=82$, disc. 22/0); Self-A vs. W2SR weak $p=1$ ($n=79$, disc. 2/2); Self-B vs. W2SR weak $p=1$ ($n=79$, disc. 2/3) — statistically indistinguishable.

collapse acknowledgment to 2.4% and 3.4% — statistically indistinguishable from W2SR weak (paired $p=1$, $\Delta \approx 0$ on $n=79$ matched pairs each) — and median CoT compresses to ∼1.3k chars regardless of budget. **Cross-substrate replication on MMLU.** A small MMLU sweep (5 STEM subjects, 8 Qs each → 40 cued/cell, $n = 35$ paired after parseable-answer attrition on the baseline side) holds the pattern: baseline ack 20.0%, W2SR weak 2.9%, Self-A 0.0%; Self-A vs. baseline paired $\Delta = -0.171$, McNemar $p = 0.031$ (discordant 6/0, one-directional); Self-A vs. W2SR weak again indistinguishable ($p = 1$). Baseline R1-7B already writes shorter CoT on MMLU (3.6k vs. 19k chars median), so MMLU compression is only ∼3× vs. GPQA's ∼14×, and the absolute ack drop is correspondingly smaller (MMLU $20.0 \to 2.9\%$, drop ∼17 pp at 3× compression; GPQA $25.0 \to 3.2\%$, drop ∼22 pp at 14×). *Less compression, smaller drop, but the drop still fires* — a within-paper dose-response from compression to ack collapse that supports the compression-dominant reading without staking it on the matched-length cuts alone. So the faithfulness collapse *is not weak-to-strong-specific, not GPQA-specific, and not contingent on extreme CoT compression*: it fires at 3× compression too, and its magnitude scales with how much compression occurred. The W2SR asymmetry is incidental; the mechanism is **SFT on terse MATH-style CoT traces in general**, which compresses CoT and suppresses cue-acknowledgment whether the teacher is weaker, stronger, or identical to the student.

6 Discussion

Math-CoT SFT does not preserve monitorability and, where it is measurable, it hurts. The two substrates are consistent: on the floored instruct student Meek's composite "monitorability" (the mean of verbosity and faithfulness scores defined in [2]) rises after distillation almost entirely via verbosity—the model *looks* more monitorable; on the reasoning student, the cue's *behavioral pull on the answer is not reduced while its verbal acknowledgment collapses*—a dissociation a compression-only account would not predict, since the model keeps being moved by the cue while no longer reporting it. The safety-relevant component thus gets *worse*, not better. The warning is sharp: *an accuracy-only certification of* ***any*** *CoT-SFT student trained on MATH-style traces—including same-strength self-distillation, not just weak-to-strong as in W2SR—can pass a model whose chain-of-thought has become markedly less revealing*, and a monitorability check that rewards elaborate-looking CoT would miss it too. The narrowness of the W2SR accuracy claim itself (in-family teacher + base student, §4) only sharpens this point: the certified guarantee is narrow on both axes.

7 Limitations

Floored instruct substrate: the instruct arm only motivates; the faithfulness claim rests on the reasoning student. **Matched-length residual is regime-dependent:** at matched mid-length, baseline and W2SR are indistinguishable; at very long lengths a real gap remains (OR $\approx$ 2.2, Fisher one-sided $p = 0.08$) but just misses significance. We claim a residual restricted to the long-trace regime, with compression as the dominant mechanism. **Single family:** the degradation is shown within the Qwen R1-distill lineage, demonstrated under weak, strong, AND same-strength (self) teachers. A same-recipe cross-family test on R1-Distill-Llama-8B (self-distillation, matched LoRA config, 8k generation budget, same 1200 MATH L3–5 problems) fails the capability gate cleanly: 50.0% $\rightarrow$ 22.5% GPQA accuracy (-27.5 pp at 0% empty completions on a verified-warm endpoint), at $\sim$10.8$\times$ CoT compression (15,420 $\rightarrow$ 1,424 chars median). This is substrate sensitivity, not infrastructure noise: the same recipe that holds Qwen-7B capability while compressing $\sim$14$\times$ over-compresses Llama-8B and craters accuracy, placing the cross-family Self-B arm in the over-compression confound regime, so we do not run its monitorability eval. Cross-family generalization of the dissociation therefore remains open: at matched recipe Llama fails the gate before the dissociation question is measurable. **Cross-substrate:** the cross-MMLU replication ($n = 35$ paired) is direction-stable and significant for the Self-A vs. baseline comparison ($p = 0.031$, 6/0 discordant) but underpowered for W2SR weak vs. baseline at this n ($p = 0.125$, 6/1); broader MMLU coverage (humanities, social sciences) is untested. **Also:** LoRA (not full SFT), MATH-L3–5 SFT data only (other domains untested); judge robustness is checked with one alternative family (`gemini-2.5-pro`), not a panel.

8 Conclusion

W2SR's accuracy gain is real but conditional (in-family teacher; genuine only on a base student), and on a reasoning student its faithfulness collapse (25% $\rightarrow$ 3% at preserved accuracy) is **not weak-to-strong-specific**: a same-strength self-distillation control reproduces the collapse at indistinguishable magnitude, and the cue's behavioral pull on the answer is *not reduced* while verbalization collapses—a dissociation that survives a within-judge rubric swap (under which it tightens: ack 9.4% $\rightarrow$ 0.0%), an out-of-family judge, and a cross-substrate (MMLU) replication. The mechanism is *compression as produced by math-CoT SFT* — at properly matched length the residual is not statistically significant, but the substrate dose-response (GPQA 14$\times$, MMLU 3$\times$) supports compression as the load-bearing mechanism rather than weak-to-strong asymmetry. A same-recipe cross-family attempt on R1-Distill-Llama-8B fails the capability gate cleanly (-27.5pp accuracy at 10.8$\times$ compression), so the dissociation's cross-family scope remains open. Certifying *any* CoT-SFT student on accuracy alone, or on how elaborate its CoT looks, is insufficient and potentially misleading; the cue-acknowledgment component of monitorability must be measured directly.

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